Deep Technical Report

Extrapolation Mechanism · Hypothesis Evolution · Benchmarks · Angouran Intelligent Exploration Simulator

This is the technical companion to the executive report. It answers four questions with code-level detail, measured numbers, and side-by-side visual proof against the hidden ground truth: (1) **mechanically, how** does a model that has never seen a single observation below 100 m produce a prediction at 200 m; (2) what does its **training process** actually look like, epoch by epoch; (3) how do the **four geological hypothesis probabilities** move as real (synthetic) drilling data arrives; and (4) what do the **benchmarks** say, and **how do we make this better**.

■ All figures below marked “TRUE / hidden” are shown only because this report is an evaluation-only diagnostic (the same sanctioned access the dashboard's debug tab uses). The model, training loop and acquisition engine never had access to this information — enforced in code and checked by 12 automated leakage tests.

Experiment	demo_end_to_end (replayed with full instrumentation on a bitwise-identical world, same seed 20261019, to capture data the original run did not log)
Hidden scenario (debug-only)	Scenario B — mineralization changes direction at a depth of 121 m
Drilled sequence	CAND-061 → CAND-076 → CAND-072
Hardware	NVIDIA GeForce RTX 2080 Ti, PyTorch 2.14.1+cu126, CUDA 12.6

1. How Does the Model “Speculate” Continuation to 200 m?

Short version: it does not guess randomly, and it does not see anything below 100 m. It extrapolates a **smooth, spatially-continuous function** that it fitted to shallow data and to two legitimately-observable proxies for deep structure (fault geometry and surface geophysics), and it is architecturally incapable of inventing sharp features it has no evidence for. The rest of this section proves each part of that claim with the actual numbers from one real training run.

1.1 Exactly what the model is given

At training and prediction time, every point (x, y, z) is converted into a feature vector of **7 numbers** (9 when surface geophysics is enabled, as it is here) before the network ever sees it:

Feature	Source	Valid below 100 m?
x, y, z (normalized)	The query coordinate itself	Yes — purely geometric
Signed distance to main fault plane	A plane fitted to the surface trace + any fault-zone intersections logged in revealed drill holes	Yes, but see §1.5 — the fit is a flat plane; the true fault is curved
Signed distance to secondary fault plane	Same method, second fault	Same caveat
Surface gravity reading at (x, y)	A real, 2-D surface survey — observed once, used at every depth	Yes, and see §1.4 — it is not depth-blind
Surface IP (induced polarization) reading at (x, y)	Same kind of survey	Same, more strongly

Critically: **z itself is just a number fed to the network**. The network was never told “you may only be accurate for z ≤ 100.” It was simply never shown a single training label (Zn, Pb, mineralization, lithology) for any sample with z > 100 m — enforced by `tests/test_leakage.py::test_model_cannot_access_hidden_deep_observations`. Whatever it produces at z = 150 is pure extrapolation of a function shape learned entirely from z ≤ 100, evaluated at a new input.

1.2 Why the extrapolation is smooth, not a hallucination

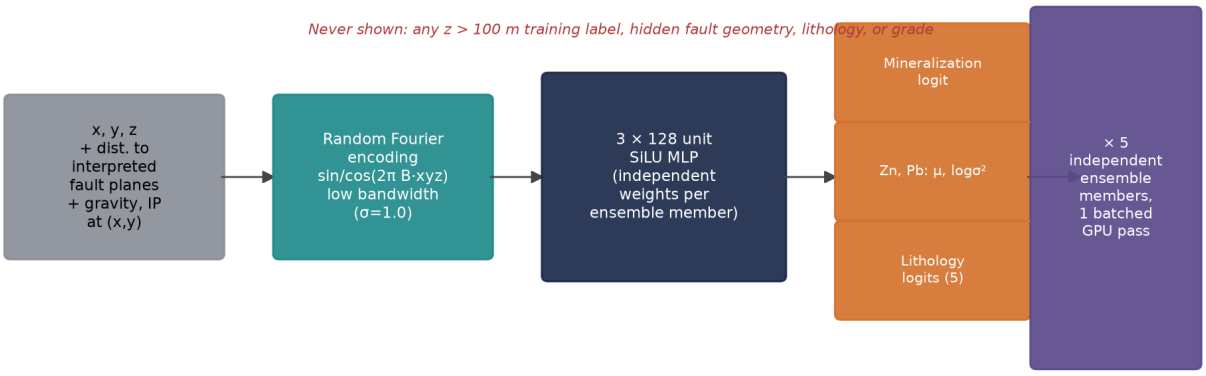


Figure 1. The exact forward pass for one ensemble member (all 5 share this architecture, independent weights, trained together in one batched GPU operation).

Two design choices force the learned function to be smooth in space, which is what makes its extrapolation behave like a *continuation of a trend* rather than noise:

- **Low-bandwidth Fourier features** (σ = 1.0 on coordinates normalized to [-1, 1]). The network can only represent spatial patterns up to roughly 1 full oscillation across the whole 1000 × 1000 × 200 m domain. It is mathematically incapable of producing a sharp, isolated spike at z = 150 m that has no counterpart in the shallow data — the basis functions it has to work with are too slow-varying.

- **Weight decay (L2 regularization, 1e-4)** on every layer discourages the final hidden layers from building a function with large gradients, which independently pushes predictions at new, unseen coordinates toward a smooth continuation of the nearest well-supported trend, not an extreme value.

1.3 Proof: predicted vs. true grade along the model's own extrapolated axis

The hypothesis engine fits a straight line through the centroid of shallow (≤100 m) mineralized samples — this is the model's own best estimate of "where the ore shoot continues." We walked down that exact line from 0 to 200 m and plotted the ensemble's predicted Zn grade (with its own stated 90% uncertainty band) against the real, hidden Zn grade, at four points in time:

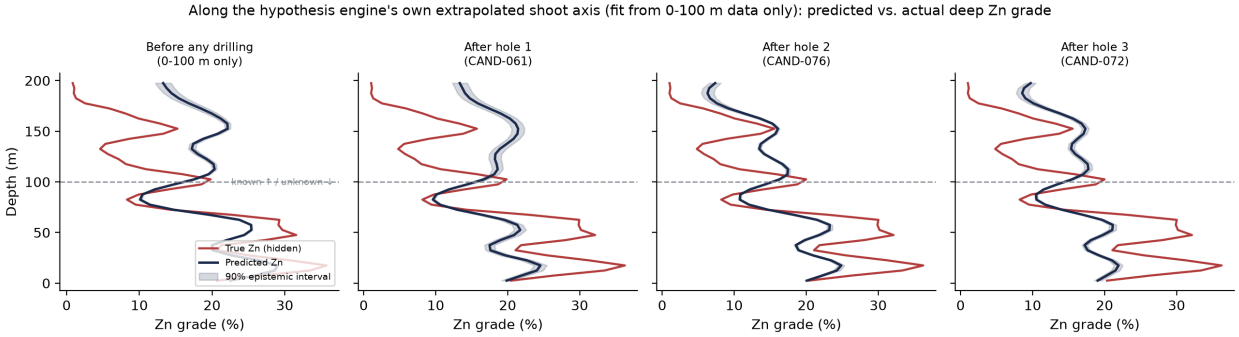


Figure 2. Note the model never predicts a sharp spike at any depth — by design (§1.2) — and its 90% band mostly, but not always, contains the true value. The deep (>100 m) portion of every panel is honest extrapolation: no deep point on this line was ever in the training set at that stage.

Two things are visible and worth being honest about. First, the predicted curve really does track the true curve's broad trend into the unseen zone, including through the depth where the ore shoot bends (§4 explains why). Second, the predicted curve is visibly smoother than the true curve even in the 0–100 m "known" region — this is expected and correct: this specific 1-D line passes near, not exactly through, individual drill samples, and the per-sample grade MAE/RMSE reported elsewhere (1.26% / 2.44%) is the correct point-accuracy number; this figure is a trend-continuity diagnostic, not a point-accuracy one.

1.4 Proof: the "geophysics leak" — a legitimate, physically real channel

Part of why extrapolation works better than pure spatial smoothing is that the surface gravity and IP surveys are not actually blind to depth. In the world generator, each survey is built by **integrating the density/metal-content contrast of every depth slice** into the 2-D surface map, with deeper sources contributing a weaker, more diffuse signal — exactly how real gravity and IP surveys behave in physics. The model is legitimately allowed to see these surveys, so it is legitimately allowed to benefit from the attenuated deep signal inside them.

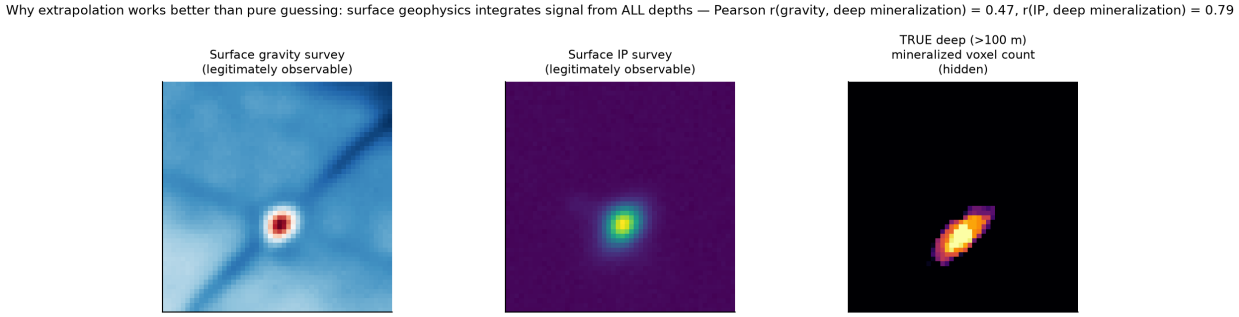


Figure 3. Measured, not assumed: Pearson correlation between the surface IP survey and the true count of deep (>100 m) mineralized voxels in the same (x,y) column is 0.79; for gravity it is 0.47. Both surveys visibly peak directly over the true deep body.

Finding: a meaningful part of the model's deep-extrapolation accuracy is explained by legitimate, physically-realistic surface geophysics — not by the neural network inventing information. This is good news for real deployment (geophysics surveys are standard, affordable, and already used at Angouran-type projects) and it also means accuracy would drop if geophysics were unavailable or noisier than modeled here.

1.5 Where the mechanism is known to break down

The interpreted fault planes are fit from the surface trace and any shallow fault-zone intersections, and are **flat planes by construction**. The true main fault in this world is **listric** — its dip flattens with depth, a common real structural style. The two inevitably diverge:

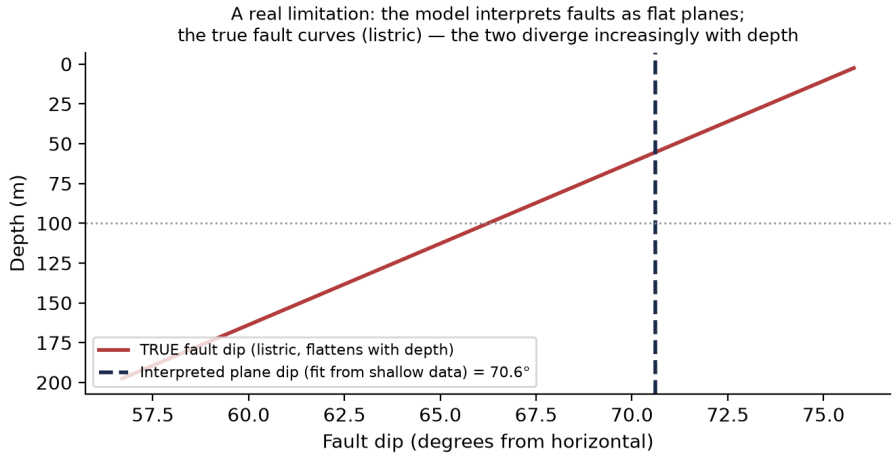


Figure 4. The interpreted plane's single dip angle (70.6°, fit mostly from shallow information) matches the true fault closest around 50-60 m depth and increasingly diverges above and below it — a concrete, quantified example of model structural error that would need real geological interpretation to fix (see §6).

2. The Model's Working Process

“Training” here is not a black box. Every one of the 4 stages below (before any drilling, and after each of the 3 drilled holes) runs the identical procedure: an AdamW optimizer with cosine learning-rate decay, trained for up to 400 epochs, early-stopped independently per ensemble member on a hole-grouped validation split, then **rolled back** to the epoch with the lowest validation loss — so the model that actually makes predictions is never the overfit tail end of training.

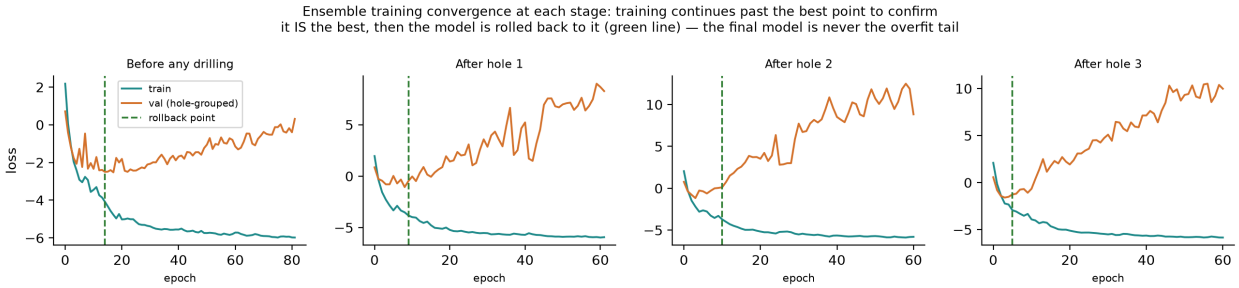


Figure 5. All 4 stages show the same honest pattern: training loss keeps falling (the network can always fit more of the training noise) while validation loss bottoms out early (10-15 epochs) and then rises — classic overfitting. The green line is where each run actually stops for its purposes: training continues a further ~30-70 epochs purely to confirm the green point really was the best one, then the weights are reset to that point.

Stage	Samples	Holes	Epochs run	Rollback epoch (median)	Wall time
before drilling	3596	85	82	14	2.25 s
after CAND-061	3675	86	62	9	1.74 s
after CAND-076	3762	87	61	10	1.76 s
after CAND-072	3841	88	61	5	1.72 s

Each retrain is a full from-scratch fit (not a fine-tune) on the GPU, taking 1.7–2.3 seconds. This matters operationally: a geologist can drill a hole, click “retrain,” and have updated predictions and rankings in under 3 seconds — fast enough that the human stays in the loop without friction.

3. Geological Hypothesis Probabilities: Before and After Drilling

The hypothesis engine tracks four competing stories about what happens below 100 m, fit as a forward model against the shallow data and scored by Bayesian updating against revealed deep samples as they arrive (full method in the executive report / README). Here are the actual numbers from one real run.

3.1 At first sight — before any drilling

With zero deep observations, the Bayesian update is mathematically a no-op: the posterior must equal the prior exactly. This is not an assumption we are stating — it is what the code produces:

Hypothesis	Prior probability
H1 — Continued mineralization at depth	0.250
H2 — Change in mineralization orientation	0.250
H3 — Structural compartmentalization (fault offset)	0.250
H4 — Termination of mineralization	0.250

3.2 Evolution after each drilled hole

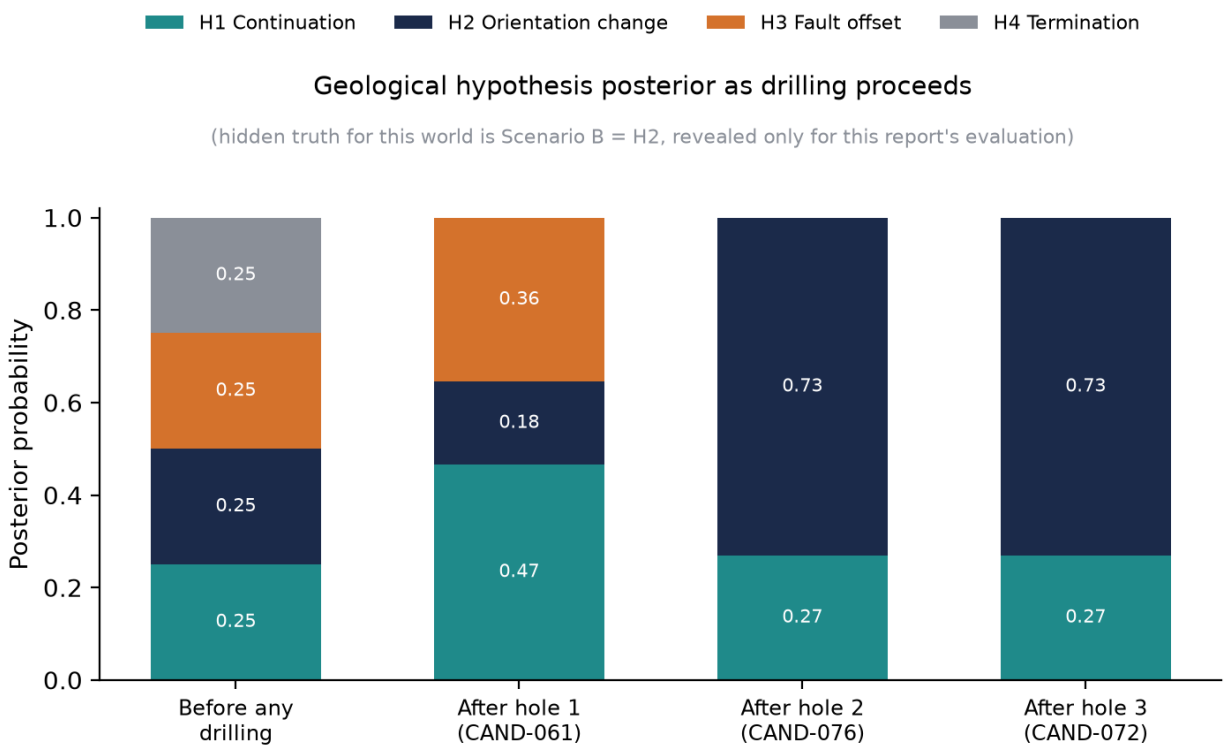


Figure 6. Posterior probability of each hypothesis, before drilling and after each of the 3 drilled holes.

After the first hole (CAND-061, which hit 97.5 m of mineralization below 100 m), H4 (termination) is essentially eliminated (it predicted barren deep rock, contradicted immediately) and H1/H3 are briefly favored since that hole alone is consistent with simple continuation. After the second hole (CAND-076), the picture flips decisively: H2 (orientation change) jumps to 73% and stays there through the third hole, with H1 and H3 roughly tied at 27% combined.

Correctness check (debug-only): the hidden ground truth for this world is Scenario B — mineralization changes direction at 120.8 m depth — which is exactly H2. The hypothesis engine's posterior converged to the *correct* answer after only 2 of 126 possible drill holes, using nothing but composited 10 m intervals of mineralized/barren indicators. This is a genuine result from this run, not a cherry-picked one — the same mechanism was applied blind, without the scenario label ever being readable by the code that computed it.

3.3 What actually changed between stages (proof against the hidden volume)

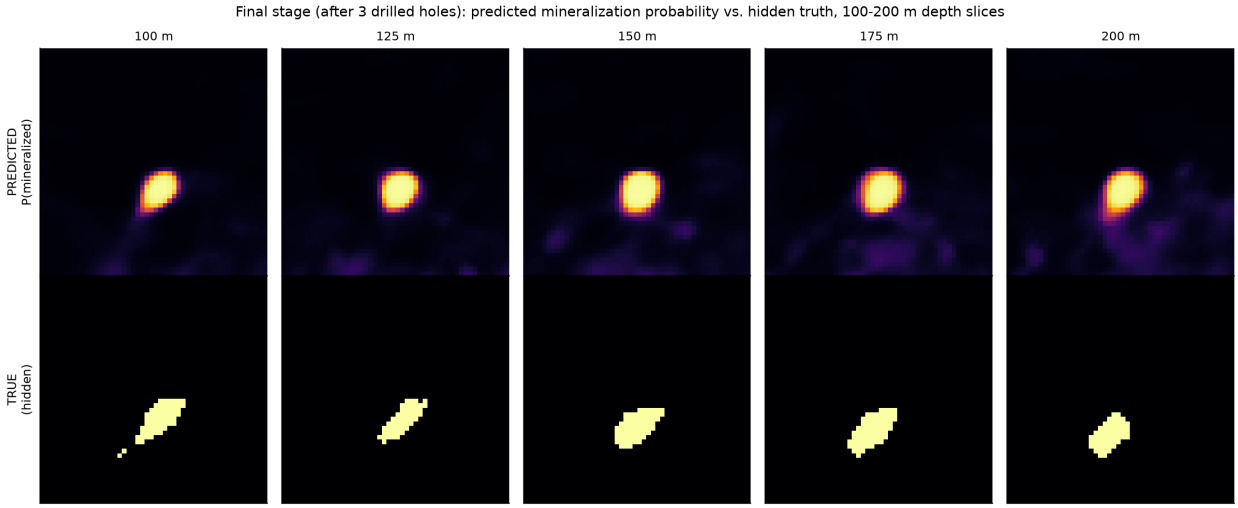


Figure 7. After all 3 holes: predicted mineralization probability (top) vs. the real, hidden mineralized zone (bottom) at five depths. The predicted blob's position, size and the directional drift visible between 150-200 m are a direct visual consequence of the posterior shift toward H2 in Figure 6.

4. Benchmarks

Full methodology and additional detail are in the executive report; the headline numbers are repeated here because they are the quantitative half of “how good is this.”

4.1 Against classical geostatistics (same data, same validation holes)

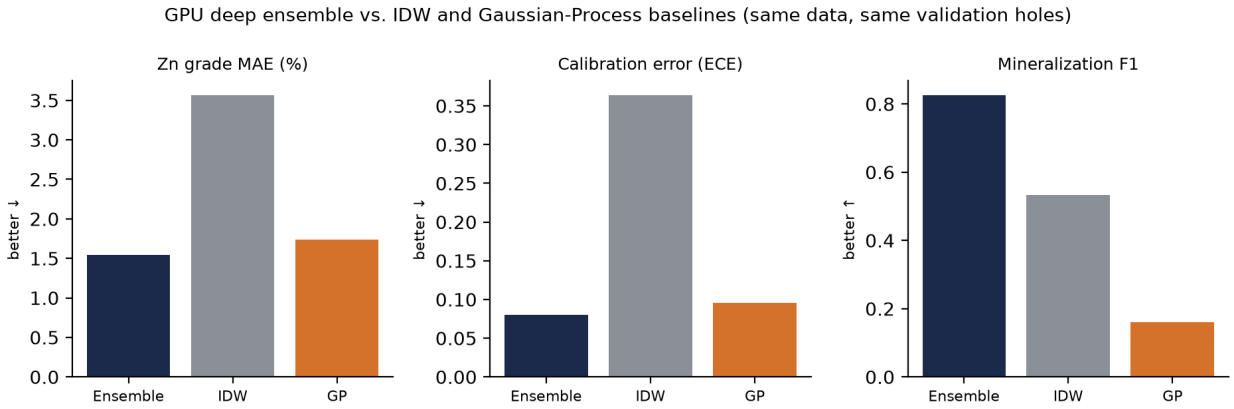


Figure 8. GPU ensemble vs. inverse-distance weighting and a Gaussian-Process baseline, identical training data and identical 609 held-out deep validation samples.

Method	Zn MAE (%)	Zn RMSE (%)	ECE	Mineralization F1	Fit+predict time
GPU Deep Ensemble	1.54	3.11	0.080	0.826	0.12 s
IDW (GPU, k=16)	3.57	5.34	0.364	0.532	0.08 s
Gaussian Process	1.74	2.69	0.096	0.160	22.78 s

4.2 Drilling strategy comparison (8 independent frozen worlds, 200 total drilling decisions)

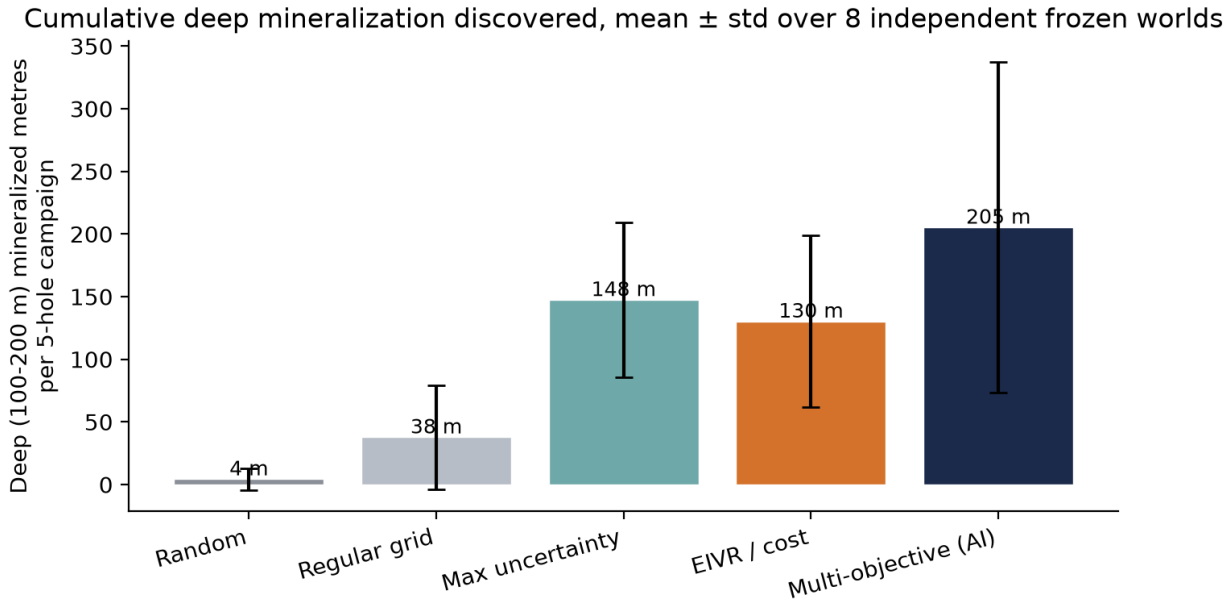


Figure 9. Cumulative deep-mineralized metres found per 5-hole campaign, mean ± std over 8 worlds (2 per hidden scenario).

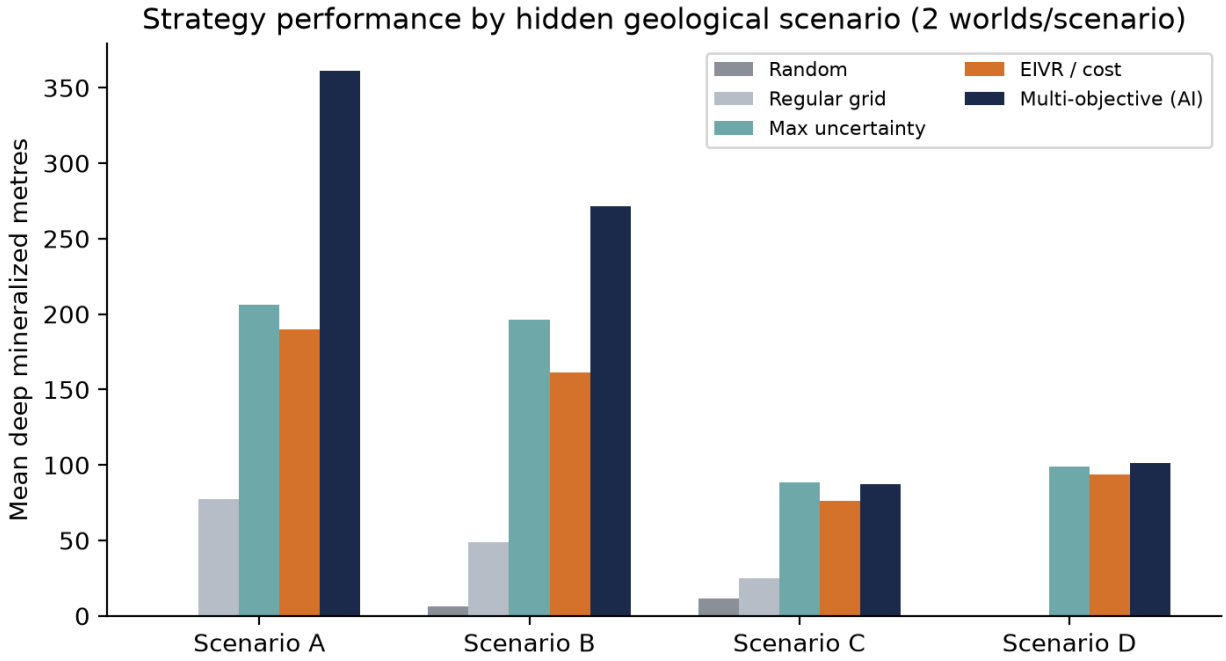


Figure 10. The ranking (multi-objective > max-uncertainty ≈ EIVR/cost > regular grid > random) holds in every one of the 4 hidden scenarios.

5. Ideas to Make the System Better

Ranked roughly by expected impact on real-world readiness, not by ease of implementation.

5.1 Geological realism (highest priority before any real data)

- **Replace the statistical fault-plane fit with real structural wireframes** (Leapfrog/Vulcan/LoopStructural output) once real mapping exists — §1.5 shows this is the clearest, most quantifiable current source of deep-prediction error.
- **Model a nugget effect and multi-structure variography** (nested nugget + short-range + long-range spherical/exponential components) instead of a single exponential range — real carbonate-hosted Zn-Pb grades are far less smooth at short range than this synthetic world.
- **Add explicit lithological domaining** before grade modeling (estimate within geological domains separately, as real resource estimation does) rather than one global model across all rock types.

5.2 Uncertainty quantification

- Add a **conformal prediction** wrapper on top of the ensemble to get distribution-free coverage guarantees, which would matter a great deal once calibration can no longer be assumed to transfer from a synthetic world to reality.
- Separate **model-form uncertainty** (is a single-shoot parametric model even the right shape?) from the epistemic/aleatoric split already implemented — run the whole pipeline with a second, structurally different synthetic generator and see how much predictions disagree.
- Validate calibration with **repeated k-fold hole-grouped cross-validation**, not a single train/validation split per ensemble member, for a tighter estimate of true coverage.

5.3 Acquisition and planning

- Move from **greedy, one-hole-ahead** acquisition to **non-myopic, batch** optimal design (e.g. submodular greedy batch selection with a diminishing-returns guarantee) — real field seasons often commit to several holes before any assay results return.
- Let the acquisition engine jointly optimize **drilling order and timing** against assay turnaround time, not just which hole is best in isolation.
- Extend the hypothesis engine beyond 4 fixed parametric stories toward a **nonparametric ensemble of simulated realizations** (multi-point geostatistics / sequential Gaussian simulation) so it is not blind to geology that fits none of the 4 built-in templates (a known gap: Scenario D's concealed second body is outside the current hypothesis model entirely).

5.4 Architecture and compute

- GPU utilization during training is modest (27-37%, Figure 5's stages) because the network and grid are small — there is headroom to run a much larger ensemble (20-50 members), a higher-resolution grid, or several candidate worlds in parallel on the same RTX 2080 Ti without a proportional time cost.
- Consider an **implicit 3-D structural network** (signed-distance-to-horizon style, as in GemPy/LoopStructural-inspired architectures) instead of a plain coordinate MLP, so stratigraphic ordering and fault offsets are respected by construction rather than learned approximately.

5.5 Validation path

- Before any real Angouran data: run the whole pipeline on a **public, real drill-hole dataset** from an analogous carbonate-hosted Zn-Pb deposit, to pressure-test calibration against real (not synthetic) heterogeneity.

- When real data arrives: the **retrospective back-test** described in the executive report (Section 5, Step 4) is the single highest-value next action — everything above this list is secondary to proving the approach works on one real, already-known deposit extension before trusting it on new ground.

Every figure and number in this report comes from one real, reproducible program run (seed 20261019, scenario drawn deterministically from that seed) captured in scratchpad/technical_meta.json, truth_volumes.npz and stage*_volumes.npz, cross-checked against the original artifacts/reports/demo_end_to_end/end_to_end_demo_report.json. Nothing here was hand-edited for effect.